Omer Kilic

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Aerospace Engineering · Computational Aerodynamics · Scientific Computing · Manufacturing & Testing · Propulsion

EDUCATION

Rutgers, The State University of New Jersey · New Brunswick, NJ · B.S. Aerospace Engineering — Expected May 2028
GPA 4.00/4.00 · Aresty Research Scholar · Richard Peters & Cecilia Santiago Peters Endowed Scholar

Bergen Community College · Paramus, NJ · A.S. Engineering Science — 2024-2025
GPA 3.90/4.00 · Phi Theta Kappa · USDA 3SP Scholar ($3,000)

Bilkent University · Ankara, Turkiye — 2017 – 2020
Undergraduate studies in Economics and Business; relocated to the United States in 2020 · Full scholarship covering tuition and stipend · National University Entrance rank: **1,085 of 2.5M** candidates

TECHNICAL SKILLS

Aerodynamics & Numerics discrete-vortex methods, multipole expansions, lumped-wake modeling, fast multipole method, unsteady wake modeling, 6-DOF store-separation trajectory analysis, flight dynamics and stability derivatives, CFD meshing and post-processing
Scientific Computing Julia, Python (NumPy, Pandas, Matplotlib, Jupyter), MATLAB, C++, Git, LaTeX/Typst
Optimization & Verification automatic differentiation, gradient-based design optimization, Monte Carlo / GCMC, surrogate and reduced-order modeling, numerical verification, error analysis and convergence study
Design & Manufacturing SolidWorks, composites manufacturing, rotor/propeller balancing, GD&T, NX/Fibersim workflow exposure, NCR/MRB
Test & Instrumentation wind-tunnel test exposure, sensor monitoring and diagnostics, CAN Bus, NEC code compliance

RESEARCH & AEROSPACE EXPERIENCE

Aresty Research Assistant · Flapping-Wing Aerodynamics Lab, Rutgers Mechanical and Aerospace Engineering Department. — 2026 – Present
PI: Prof. Mitsunori Denda · Bio-Inspired Micro Aerial Vehicles · insectflight.rutgers.edu

- Derived closed-form multipole moments of polygonal vortex rings up to the octupole, with shift and combination rules and an a priori far-field error bound; co-author, with Prof. M. Denda and A. Malhotra, of the laboratory notes "Multipole Expansion for the Vector Potential" (2026).
- Formulated and implemented the 3-D lumped-wake rule: one equivalent vortex ring per group of far-wake rings, carrying its impulse, placed by its quadrupole and shaped by its second moment of area; a switch in the group's MATLAB 3-D wake solver, verified against a reference implementation.
- Ran the 80-step direct-versus-multipole comparison, the first lumped-wake runs and a 48-run time-step, mesh and core-radius sensitivity study; traced the late-time force to the old wake's deformation. First-author manuscript in preparation.
- Built a Julia library of ring moments and a multipole far field (**40 automated test groups, 274 checks**); replaced the solver's velocity kernel with an exact vectorized one, cutting a 15-step run from **1,783 s to 50 s**. Wrote the group's FMM and Julia tutorials; built and maintain the group hub.

Candidate Engineer · SKY Global Turk Internship Program · Turkish Aerospace Industries (TUSAŞ) · Ankara, Turkiye — Aug – Sep 2026
Turkiye Internship Program (YTB) · selected via technical interview · UAV Manufacturing & KAAN Aerodynamics

- Ran KAAN store-separation aerodynamic studies end to end and reported the results: geometry preparation, meshing, solver execution, 6-DOF trajectory analysis, post-processing; observed scaled-store wind-tunnel tests across multiple configurations.
- Delivered a first-principles flight-dynamics analysis of the KAAN fighter: derived the 6-DOF Newton–Euler equations of motion, stability derivatives and spin behavior and documented how the aerodynamic database and surrogate models are built.
- Delivered a technical report on T129 ATAK rotor-blade manufacturing and balancing to evaluate production of AKSUNGUR UAV propeller blades (master tooling, tolerances, CG control), including gram-scale tip trim-mass corrections and their moment-arm effect on balance and vibration.
- Documented the design-to-production data chain (composite manufacturing, Teamcenter routing, NX/Fibersim ply data, NCR/MRB) across the AKSUNGUR, ANKA-3, Super Şimşek and AYAZ programs, analyzing process consistency

Propulsion Lead · Rutgers Rocket Propulsion Lab (AIAA) · New Brunswick, NJ — Jan 2025 – Present
Student-built solid rocket motors · motor design, manufacturing, static testing, data analysis

- Student-built 98 mm solid rocket motor, casing to static fire: single-point threaded the 6061-T6 casing and proof-tested it hydrostatically to 1,200 psi; sized the throat in openMotor; redesigned the phenolic/graphite nozzle from sectioned char measurements (safety factors 17.3 bearing, 12.2 carrier, 2.8 insert); two static fires on 13 Dec 2025 at ≈5,380 and 5,100 N·s (L/M class boundary), ≈1,190 N peak thrust, ≈5 s burn.
- Re-reduced the thrust-stand logs from the raw samples (zero-offset correction worth 8% of total impulse) and fitted the burn-rate law $a \cdot P^n$ from the firing pressure data by Nakka's method ($n \approx 0.67$, MATLAB and Excel in agreement); built the team's openMotor propellant and nozzle models.
- 2026–27: competition grains and a 75 mm head-end-ignition sustainer for Jersey Devil, the team's first flight on a motor of its own.

Research Intern · BCC-NJIT Summer Research Program · Laboratory for Materials Interfaces (Prof. Gennady Gor), Otto H. York Department of Chemical and Materials Engineering · New Jersey Institute of Technology · Newark, NJ — Summer 2025

- Ran Grand Canonical Monte Carlo simulations of helium confined in zirconium nanovoids at 573 K and 973 K (Mezcal), establishing thermodynamic baselines against irradiation literature; built Python/Jupyter workflows automating runs, equilibrium assessment and density/pressure-vs-μ analysis.

PROJECTS & ADDITIONAL EXPERIENCE

Founder & Electrical Technician · Hi Tech Electrical Contractors Inc. · Palisades Park, NJ — 2020 – Present
Electrician since 2020; incorporated the company in 2023 and operate it while studying full time; perform panel upgrades, branch-circuit wiring, NEC-compliant installations, and fault diagnostics. Home Depot Pro Referral partner with **22 verified five-star reviews** (5.0/5.0). [Link].

Project Member · Electric Vehicle Conversion · Bergen Community College STEM Research Center · Paramus, NJ — 2024 – 2025
Converted a 1998 Chevrolet to an electric drivetrain; developed a CAN Bus sensor monitoring and fault-diagnostic system.